

Fundamentals of Classical Mechanics

Lecture Notes - Chapter 4: Newton's First Law and Inertia

4.1 The Concept of Net Force and Constant Motion

One of the most persistent misconceptions in introductory physics is the belief that sustained motion requires a continuous net force (the medieval impetus theory). In reality, Newton's First Law establishes that an object at rest remains at rest, and an object in motion continues with constant velocity in a straight line, unless acted upon by a non-zero net external force.

In frictionless deep space, a spacecraft traveling at 20,000 km/h requires exactly ZERO continuous engine thrust to maintain its velocity. Net force causes acceleration (a change in velocity), not velocity itself.

4.2 Inertia is Mass, Not an Opposing Force

Students frequently treat inertia as if it were an active counter-force that pushes back against motion or drains kinetic energy. This is physically incorrect.

Inertia is an intrinsic, passive property of matter quantified by mass. It represents an object's resistance to changes in its state of motion (acceleration), as formulated in Newton's Second Law: $F_{\text{net}} = m \cdot a$.

A 10,000 kg spacecraft drifting through a vacuum does not experience an internal inertia resistance that must be overcome by propulsion. Both a 1 kg sphere and a 100 kg sphere coasting at 5 m/s experience exactly zero net external force.

4.3 Equilibrium in Moving Reference Frames

Dynamic equilibrium occurs when an object moves with constant velocity (speed and direction are invariant). Under dynamic equilibrium, the vector sum of all external forces is zero: $\Sigma F = 0$.

Consider an elevator cab moving vertically upward at a constant speed of 3.0 m/s. Because the speed is constant, acceleration is exactly zero ($a = 0$). Therefore, the upward cable tension (T) must exactly equal the downward gravitational weight ($W = mg$). Upward motion does not require cable tension to exceed weight.